Enhanced Image Classification via Hybridizing Quantum Dynamics with Classical Neural Networks

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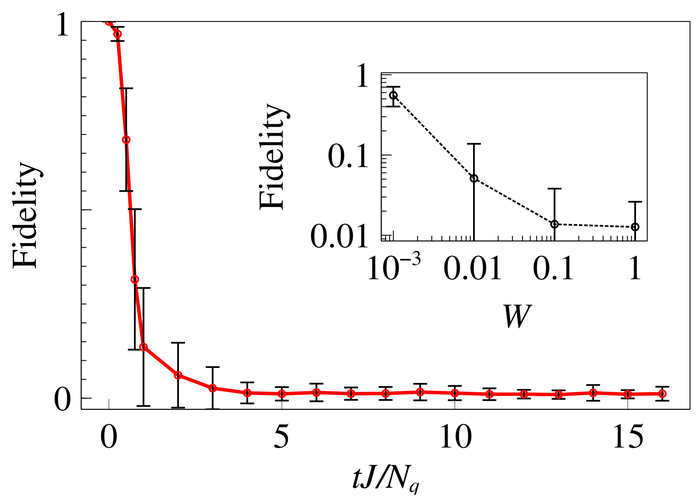
About our group

Introduction

- **Images classification:** 4 datasets about remote sensing, medical imaging, handwritten digits, and fashion.
- **A hybrid protocol:** A simple classical neural network + Hamiltonian time evolution of a spin chain.
- **Non-equilibrium dynamics:** Few qubits significantly enhance the classification accuracy.

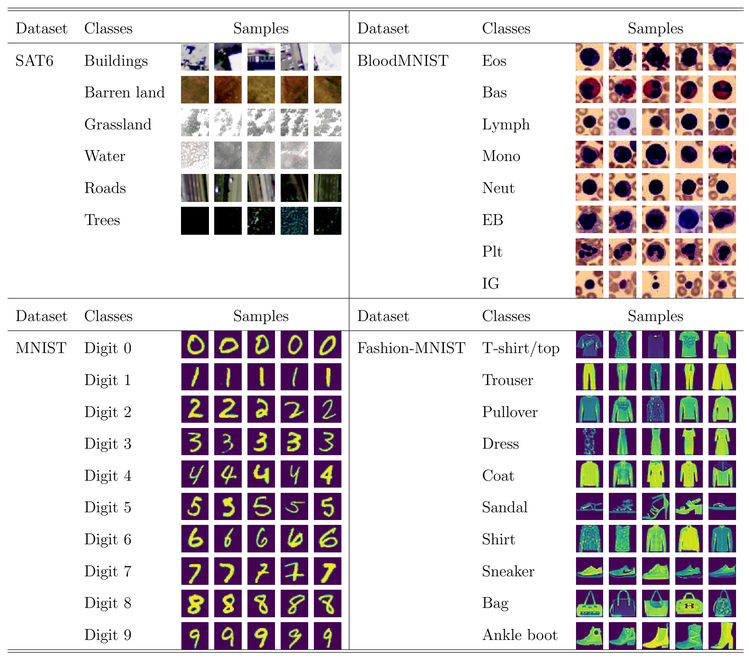
Key quantum dynamics

Identical initial states + Hamiltonian time evolution under different parameters = distinguishable final states.



Methods

Datasets



Hamiltonian for N_q qubits

$$H = J\sum_{j=1}^{N_q-1}\sigma_j^x\sigma_{j+1}^x + \sum_{j=1}^{N_q} h_j\sigma_j^z,$$

Observables for N_c classes

$$O_\alpha = \frac{1}{N_S}\sum_{k=1}^{N_S}|\Psi_k^{(\alpha)}(t)\rangle\langle\Psi_k^{(\alpha)}(t)|, \alpha = 1,2,\ldots,N_c.$$

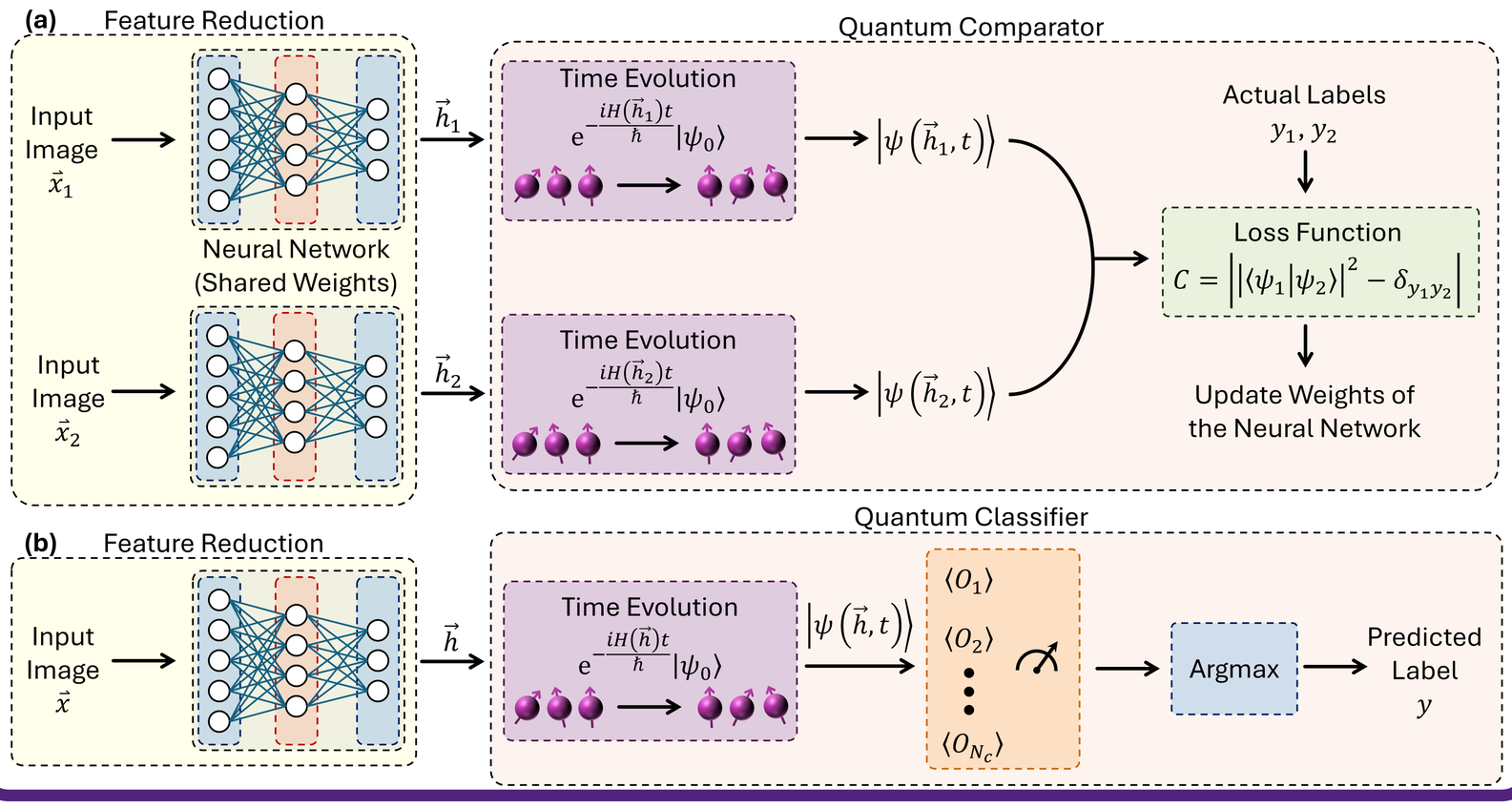
Two kinds of measurement

1. **Exact expectation:** $\langle O_\alpha\rangle = \langle\Psi|O_\alpha|\Psi\rangle$.
2. **Shadow tomography:** $\langle O_\alpha\rangle = \frac{1}{M}\sum_{m=1}^{M}\mathrm{Tr}\left(\rho^{(m)}O_\alpha\right)$.

Loss function

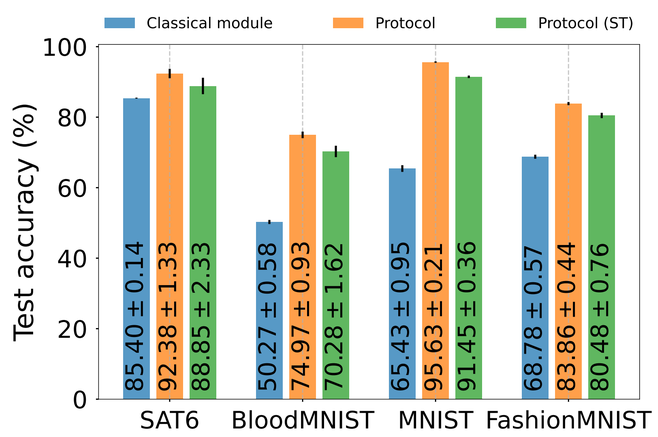
$$C = \frac{1}{N_{trn}}\sum_{\{x_1,x_2\}}\left||\langle\Psi_1(t)|\Psi_2(t)\rangle|^2 - \delta y_1, y_2\right|.$$

Protocol

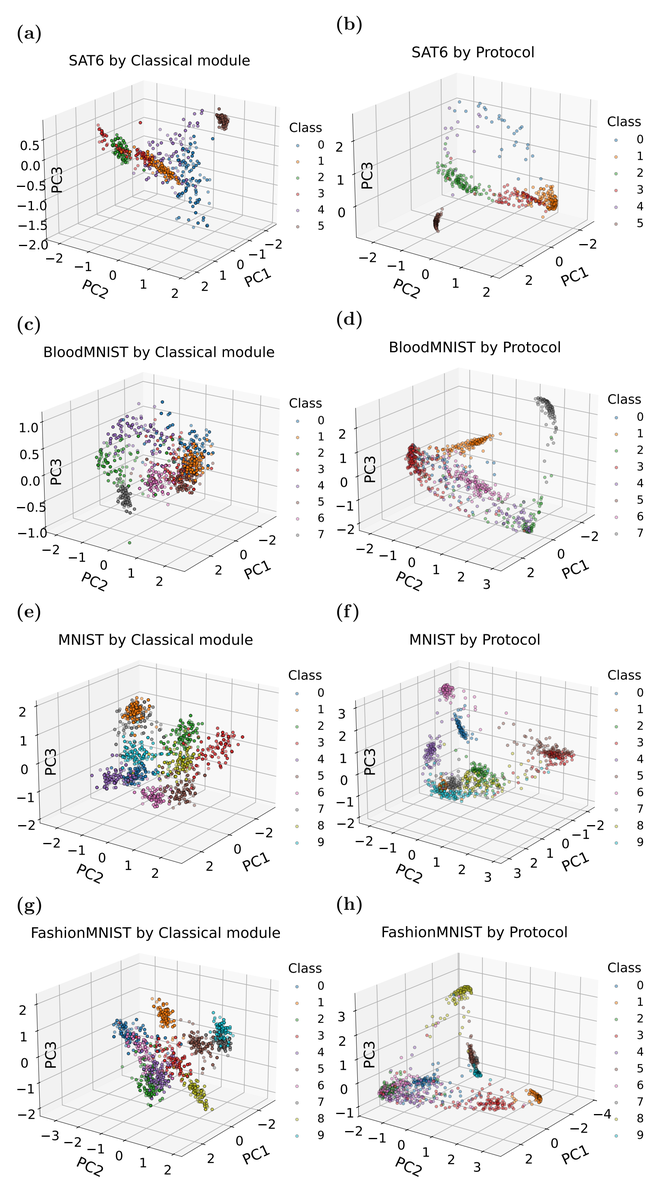


The Role of Quantum Dynamics

Test classification accuracy comparison

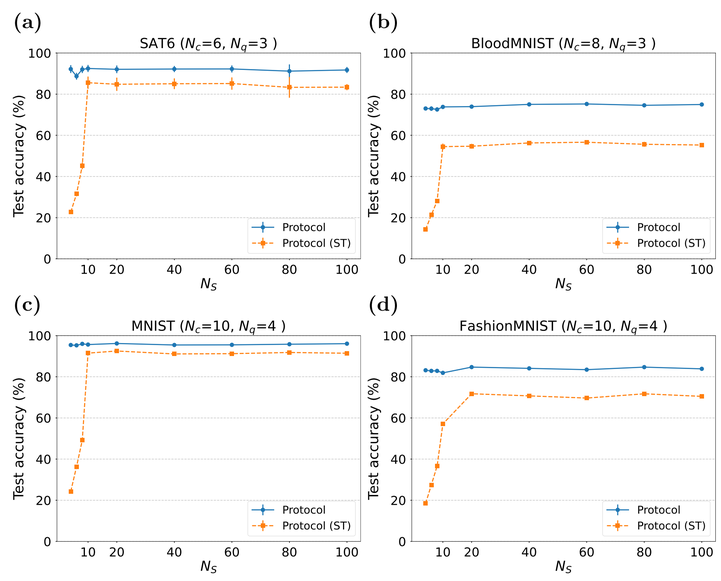


Distinguishability comparison by the PCA

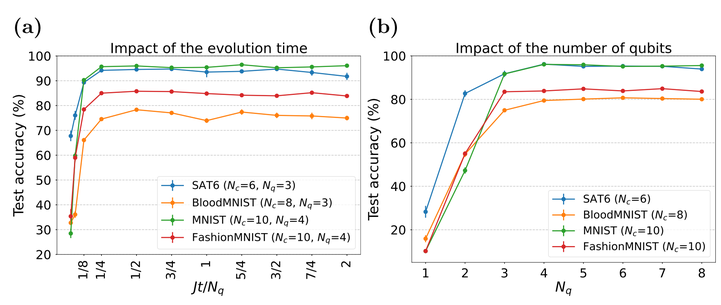


Parameter Study

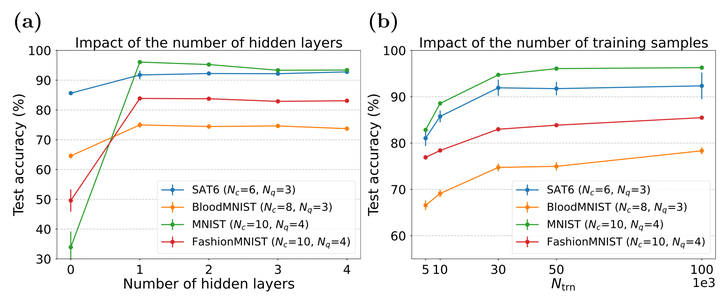
Impact of No. sample for measurement



Impact of evolution time & No. qubits



Impact of No. hidden layers & No. train samples



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